Highlights

Domain-Validity-Gated Metamorphic Testing of Scientific ML Surrogates

Meng Li, Xiaohua Yang, Jie Liu, Shiyu Yan

- A domain-validity rubric screens candidate MRs for SciML surrogate testing.
- Each MR tolerance is gated by the operator's own $O(h)$ numerical floor.
- MR cards convert retained candidates into auditable executable test assets.
- A fail-closed claim ledger ties every reported number to a tracked artifact.
- Across K=6 checkpoints, a seeded fault can hide in its own symmetry.

Domain-Validity-Gated Metamorphic Testing of Scientific ML Surrogates

Meng Li[a,b,c,*], Xiaohua Yang[a,b,c], Jie Liu[a,b,c], Shiyu Yan[a,b,c]

[a] *School of Computing, University of South China, Hengyang, China*
[b] *Hunan Engineering Research Center of Software Evaluation and Testing for Intellectual Equipment, Hengyang, China*
[c] *CNNC Key Laboratory on High Trusted Computing, Hengyang, China*

Abstract

Context: Scientific machine-learning (SciML) surrogates approximate expensive physical simulations, but exact expected outputs for arbitrary inputs are unavailable — the classical oracle problem. Metamorphic testing addresses this through relations across executions, yet a candidate relation is not automatically valid: its preconditions, output mapping, and the numerical floor of the scoring operator determine whether a violation is meaningful.

Objective: We investigate how candidate metamorphic relations (MRs) can be screened for *domain validity* and operationalized as executable, oracle-free test assets for SciML surrogates, making the step from candidate to executable asset auditable and explicit about the physical, numerical, and software conditions each relation requires.

Method: We propose (i) a domain-validity rubric admitting a candidate only when its tolerance dominates the operator's numerical floor and its preconditions hold; (ii) an MR-card / executable-asset format recording source cases, transformations, mappings, metrics, tolerances, exclusion rules, and relation-level verdicts; and (iii) a case-study protocol on MeshGraphNets cylinder-flow surrogates; a claim ledger binds every result to a tracked artifact.

Results: Across $K = 6$ checkpoints (four seed replicas, one wider and one deeper variant), node-permutation equivariance holds to machine preci-

* Corresponding author.
Email addresses: `mlemon@usc.edu.cn` (Meng Li), `xiaohua1963@foxmail.com` (Xiaohua Yang)

sion; an approximate mirror-y OOD-stress probe fails on every checkpoint with bootstrap 95% CI on median relative L_2 of $[0.74, 0.80]$, about 35× the in-distribution rollout error (CI $[0.0217, 0.0224]$); on a synthetic mesh where exact mirror-y is admitted, the surrogate still violates it ($L_2 = 1.10$), removing the out-of-relation-domain objection. The P1 discrete-divergence operator underwriting the continuity-MR admissibility predicate is empirically $O(h)$ (log–log slope 0.988, $R^2 = 1.000$). As fault detectors against a 10-mutant catalogue, the MRs catch 5 of 10 faults and localize them by class, with Wilson 95% CIs across the roster.

Conclusion: Within-architecture-family evidence supports a validity-aware bridge from candidate MRs to auditable SciML test assets, separating model-level violations from out-of-domain applications. The evidence is within one MeshGraphNets family on cylinder flow; broader generalization is future work.

Keywords: Metamorphic testing, metamorphic relation identification,
oracle problem, scientific machine learning, neural surrogates,
MeshGraphNets, domain-validity rubric

---

1. Introduction

Scientific computing increasingly uses learned surrogates to reduce the cost of repeated numerical simulation. Mesh-based neural simulators such as MeshGraphNets learn to predict physical fields on irregular meshes (Pfaff et al., 2021). These models approximate high-cost solvers but introduce a familiar software testing problem: for many candidate inputs there is no cheap exact expected output against which the program can be checked—the test oracle problem (Barr et al., 2015).

The common response is to evaluate on held-out trajectories and report rollout error. This is necessary but incomplete: a model may have acceptable average error while failing to preserve relations that should hold under physically meaningful transformations. For a user deploying a SciML surrogate outside a narrow validation set, the question is not only "How accurate is the model?" but also "Under which transformations does the system stop respecting relations it should respect?"

Metamorphic testing (MT) addresses the oracle problem by checking relations among multiple executions (Chen et al., 1998; Segura et al., 2016). Conservation laws, symmetry relations, nondimensional similarity, and continuity conditions all suggest possible relations among executions (Kanewala and Bieman, 2019; Lin et al., 2020; Olsen et al., 2019). The difficulty is deciding which candidates are valid and executable for a particular SciML program: for cylinder flow, mirror symmetry depends on geometry and boundary labels; a divergence check depends on a discrete operator, mesh weights, and boundary treatment. Treating such transformations as automatically valid MRs hides the assumptions that determine whether a violation is meaningful.

The remaining problem addressed here is therefore narrower: how candidate MRs are screened for domain validity, turned into executable test assets, and reported through relation-level verdicts that distinguish SUT inconsistency from out-of-relation-domain cases and numerical tolerance effects.

This paper treats MR identification for SciML as a validity-gated testing problem. Physical knowledge, expert reasoning, LLM-assisted candidate lists, and NOETHER-style pattern organization (Li et al., 2026) can all suggest candidate relations, but a candidate is not yet an executable MR: it must first state the physical basis, transformation preconditions, boundary-condition compatibility, output mapping, metric, tolerance rationale, exclusion rule, and verdict interpretation.

Two ideas organize this treatment. First, a candidate is *admissible* only when numerically decidable: its verdict tolerance must dominate the intrinsic error floor of the measuring operator—machine precision for an exact relation, the mapping floor for an approximate geometric relation, or the discretization floor for a continuity relation. Second, a relation-level verdict is read in two dimensions: violation magnitude and domain-violation magnitude, so a model inconsistency is not confused with a relation applied outside its domain. We refer to this as a domain-admissibility-gated, relation-indexed approach to SciML OOD validation. The case study is scoped to MeshGraphNets-family cylinder-flow surrogates; full cross-SUT evaluation remains blocked.

1.1. Research questions

The main research question is:

RQ0. How can candidate metamorphic relations for scientific machine-learning systems be screened for domain validity and converted into executable oracle-free test assets without relying on exact per-sample expected outputs?

We decompose this into four questions:

RQ1 – Validity. How can a domain-validity rubric distinguish physically meaningful candidate MRs from transformations that are executable but invalid, underspecified, or outside the relation's domain?

RQ2 – Operationalization. How can retained candidates be represented as MR cards and executable assets with source cases, follow-up transformations, metrics, thresholds, exclusions, and verdict rules?

RQ3 – Verdict. How can relation-level verdicts distinguish pass, fail, skip, out-of-relation-domain, numerical tolerance issue, and inconclusive outcomes?

RQ4 – Case-study evidence. In a MeshGraphNets-family cylinder-flow case study, what evidence does the rubric-gated asset workflow add relative to expert MR design, LLM-assisted candidate generation, generic MR-generation scope contrasts, and rollout-accuracy diagnostics? In the current evidence only the rollout-accuracy comparator is answered; the expert-MR, LLM-candidate, and generic-MR comparators remain planned and are blocked pending their artifacts.

1.2. Contributions

This paper makes three scoped contributions, each repositioned narrowly with respect to the closest prior identified in Section 2.4.

First, a domain-validity rubric and MR-card executable-asset workflow screens candidate MRs and converts retained ones into auditable executable test assets recording physical basis, transformation preconditions, output mapping, metric, tolerance, and exclusion rules. We ground the tolerance floor in the *intrinsic error floor of the discrete measurement operator* (theoretically $O(h)$ for a P1 divergence operator on a triangular mesh); in this study the floor magnitude is estimated from the reference field rather than a closed-form bound. To our knowledge, grounding an MR tolerance in the measurement operator's own error floor is new in the SciML setting.

Second, a relation-level verdict and ledger scheme reads verdicts in two dimensions (relation-violation against domain-violation magnitude). This instantiates, for physics-governed SciML, the constraint-architecture pattern of Duque-Torres et al. (Duque-Torres et al., 2023a,c) and MetaTrimmer (Duque-Torres et al., 2023b), adding a *typed classification* of domain-inadmissibility sub-dimensions drawn from PDE preconditions, geometry, boundary conditions, and operator admissibility.

Third, the paper's own MRs are used as fault detectors against an independently re-implemented 10-mutant seeded-fault catalogue, reporting by-class localization: continuity to boundary-condition/normalization faults; symmetry to physical-channel/adjacency faults; node-permutation to none. This is suggestive evidence for a typed MR-class-to-fault-class mapping, not a validated localization model.

The evidence comprises three scoped relation pilots, a same-SUT rollout-accuracy comparator, an exact mirror-y test on a provably symmetric synthetic mesh, and replication across K=6 checkpoints; expert MR design, generic MR-generation contrasts, LLM-assisted candidate generation, and cross-SUT comparisons remain protocol commitments until matched artifacts exist.

2. Background and Related Work

2.1. Mesh-based neural simulation

Mesh-based neural simulators learn dynamics on graph or mesh representations of physical systems. MeshGraphNets encodes mesh nodes and edges, propagates information by message passing, and predicts future physical fields through autoregressive rollout (Pfaff et al., 2021). The cylinder-flow benchmark exposes the distinction between data-driven accuracy and physical relation preservation. In this paper, MeshGraphNets-family systems are treated as software systems under test: we ask how outputs behave under controlled input transformations and whether necessary relations hold within explicit tolerances.

2.2. Metamorphic testing and the oracle problem

Metamorphic testing (MT) addresses programs lacking a per-input oracle (Chen et al., 1998; Segura et al., 2016; Chen et al., 2018): instead of checking each output against an expected value, MT checks a relation among outputs of a source and follow-up inputs. This framing has been applied to scientific software, simulations, and ML systems (Xie et al., 2011; Kanewala and Bieman, 2019; Olsen et al., 2019). For SciML surrogates the oracle problem is especially visible in OOD settings, where a trusted solver may be too expensive to run for every transformed case. SciML MRs cannot be treated as generic input perturbations: every transformation must respect governing equations, boundary conditions, mesh representation, and numerical tolerance, making MR correctness a central validity question.

2.3. MR identification for scientific and simulation software

MRs can be elicited from monotonicity, conservation, scaling, and domain-specific expectations (Kanewala and Bieman, 2019; Lin et al., 2020; Olsen et al., 2019; Raunak et al., 2021). Predictive MR identification has been pursued with graph-kernel learning over scientific source code (Kanewala et al., 2016), treating MR discovery itself as a learning task. Data-driven search can also discover candidate transformations in complex physical software (Hiremath et al., 2021), and design assumptions can be encoded as MRs for CPS testing under control–plant invariants (Mandrioli et al., 2025). The gap is that candidate identification is insufficient for SciML: without a domain-of-validity record a failed test is ambiguous—the relation may have been invalid under the chosen boundary conditions, the transformation may have left the physical regime, the tolerance may not have been numerically justified, or the SUT may have violated a genuine constraint. This paper makes that distinction explicit.

2.4. Physics-based MT for learned scientific simulators

Three contemporary works are the closest prior. **Reichert et al. (2024)** (Reichert et al., 2024) applied physics-derived MRs to a trained LSTM hydrologic surrogate, stratifying pass/fail by basin elevation and informally filtering basins where forcing uncertainty dominated the response. We formalise these practices as an explicit admissibility predicate and a two-dimensional verdict, and extend them to mesh-based neural fluid surrogates whose MR validity depends on geometry and discrete operators.

Eniser et al. (2022) (Eniser et al., 2022) introduced *relaxations*—numerical tolerances embedded in MR oracles—for stochastic RL policy testing. We extend this calibrated-tolerance principle to deterministic surrogates by grounding the tolerance floor in the intrinsic error floor of the discrete measurement operator (theoretically $O(h)$ for a P1 divergence operator); the floor magnitude is estimated empirically here and a closed-form bound is future work.

The 2023 violation-attribution cluster—Duque-Torres et al. (Duque-Torres et al., 2023a,c) and MetaTrimmer (Duque-Torres et al., 2023b)—addressed the bug-vs-inapplicability separation with explicit MR constraints and automated constraint derivation. Our two-dimensional verdict instantiates this pattern for physics-governed SciML, adding a *typed classification of domain-inadmissibility sub-dimensions* drawn from PDE preconditions, geometry, boundary conditions, and operator admissibility. The element none of these three works addresses is the seeded-fault MR-as-detector with by-class localization of Section 5.3.

PINN training itself exhibits gradient-flow pathologies that bias trained models away from the governing residual (Wang et al., 2021; Krishnapriyan et al., 2021), sharpening the case for relation-level validation that does not assume a converged surrogate.

2.5. SciML V&V, residuals, UQ, and failure modes

SciML reliability research has developed complementary tools: physics residuals, uncertainty quantification, conformal prediction, certified error bounds, and failure-mode analysis (Raissi et al., 2019; Karniadakis et al., 2021; Li et al., 2021; Krishnapriyan et al., 2021; Gopakumar et al., 2025). A residual or equivariance error becomes part of an executable MR only when paired with a source case, a follow-up transformation, a tolerance rule, and a verdict interpretation. We use SciML diagnostics as possible relation measurements; MT supplies the multi-execution oracle-free structure.

2.6. Hybrid ML-solver trust regions

Hybrid ML-solver frameworks use residual thresholds or error estimates to decide when a learned component should be trusted (Baral et al., 2026; Wang et al., 2025). Such systems operationalize runtime trust regions, whereas the present study acts offline: physically derived transformations are applied before deployment to estimate where relation violations occur, producing an evidence structure that can inform later deployment decisions.

The distinction from residual- and uncertainty-based trust estimation is deliberate. UQ, conformal prediction, and residual-threshold trust regions locate unreliable behavior passively from observed inputs. The present method acts in relation space: it applies a controlled transformation and reports which necessary relation breaks under it, indexed by that transformation. The two-dimensional verdict separates a model-level violation from an out-of-domain application—a separation a scalar accuracy or residual magnitude does not by itself provide. Section 5.3 gives the concrete instance: a surrogate accurate in-distribution (median one-step relative L2 0.0216) still violates mirror-y equivariance by roughly an order of magnitude, so accuracy does not bound the relation violation.

2.7. What is new and what is not new

This paper does not claim that metamorphic testing, MR identification, scientific-software MT, residual diagnostics, uncertainty quantification, LLM candidate generation, or NOETHER-style candidate organization is new. The narrower claim is that SciML MR identification should be treated as a domain-validity problem: a candidate relation becomes useful only after its physical basis, transformation preconditions, output mapping, tolerance, exclusion rule, executable artifact, and relation-level verdict are recorded. In particular, physics-based MRs for learned field predictors are an active area; we do not claim a first-or-only result there, only that the currently verified record is not strong enough to make the validity-gating workflow redundant.

What is new is the evidence-gated conversion from candidate relation to executable SciML MR asset—a workflow that makes the validity boundary inspectable so that a candidate can be retained, rejected, downgraded to OOD-stress, or deferred instead of being silently treated as a valid oracle. The structural novelty is two organizing devices: an admissibility gate that ties a relation's tolerance to the numerical error floor of its own measurement, and a two-dimensional verdict that separates a model violation from an out-of-domain application. The admissibility gate is fully operational in the case study; the verdict's domain-violation axis is so far only qualitatively operationalized, making the two-dimensional verdict an interpretive structure pending a calibrated domain-violation score. A third empirically distinct element, examined in Section 5.3, is the seeded-fault by-class localization: the MRs map by class to fault class (continuity to boundary/scale faults, symmetry to physical-channel/adjacency faults), an element none of the three closest prior works addresses.

3. Method

3.1. Overview

The proposed method is a five-stage workflow: (1) identify candidate relation sources; (2) organize candidates using declared source categories; (3) screen with a domain-validity rubric; (4) convert retained relations into executable MR assets; (5) execute assets and record relation-level verdicts. Candidate generation is deliberately separated from validity judgment:

NOETHER-style patterns may help organize candidates but do not certify that a relation is physically valid for a particular SUT. Figure 1 summarizes the workflow.

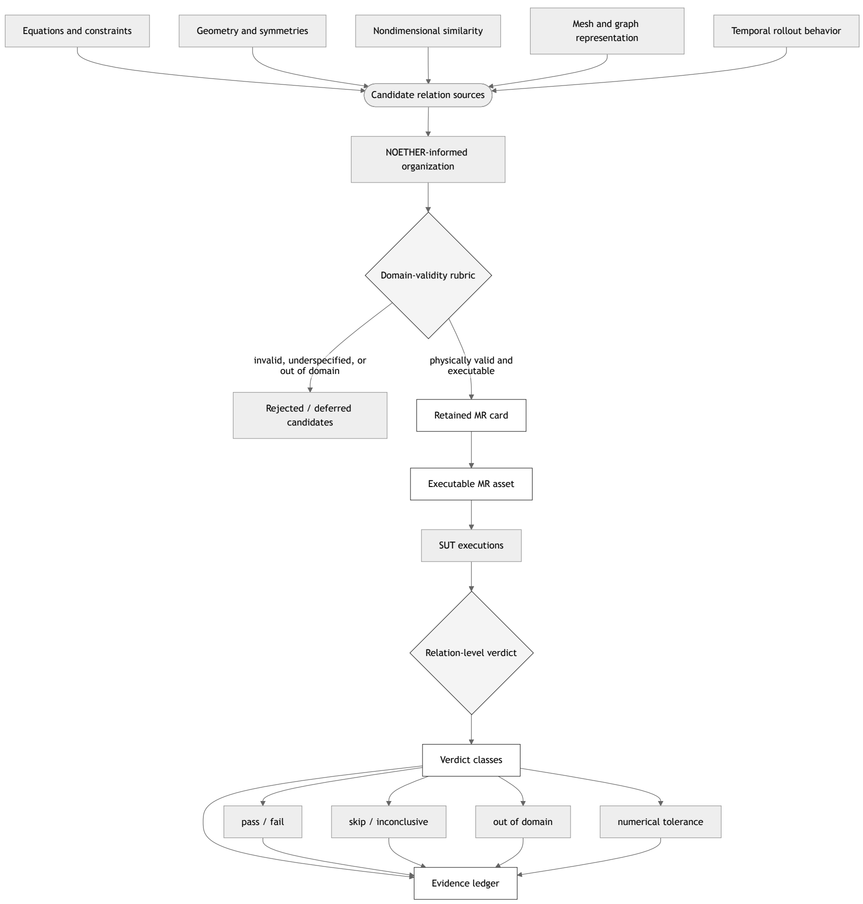

Figure 1: Validity-gated workflow for converting physically motivated candidate relations into executable, oracle-free MR assets and relation-level verdicts. The rubric and the verdict are explicit gates (diamonds); rejected candidates and out-of-relation-domain cases are recorded as evidence rather than discarded silently.

3.2. Candidate relation sources

For cylinder-flow surrogates, candidates may come from six sources: (1) **physical equations and constraints** (incompressible continuity, boundary behavior); (2) **geometric and symmetry assumptions** (mirroring, rigid transforms, coordinate-frame changes—only under compatible geometry and boundary labels); (3) **nondimensional similarity** (Reynolds/Strouhal relations, within applicable regimes); (4) **mesh and graph representation** (node permutations, face ordering, edge encoding); (5) **temporal rollout behavior** (determinism, prefix consistency—implementation sanity checks, not physical laws); (6) **cross-implementation comparison** (method-comparison checks only when units, meshes, rollout horizons, and checkpoints are comparable).

3.3. Domain-validity rubric

A candidate relation is an *admissible MR* when four conditions hold together: (i) it has a physical or software basis; (ii) its transformation preconditions are satisfied; (iii) boundary conditions and output mapping remain compatible after the transformation; and (iv) it is numerically decidable—the verdict tolerance dominates the intrinsic error floor of the measuring operator. None of the four is optional: each is a gate that can reject or defer a candidate. The six rubric criteria in Table 1 are the auditable form of these four conditions.

The rubric yields one of four decisions: retained as executable MR; retained as OOD stress relation; rejected; or deferred pending missing evidence. The defer and OOD-stress decisions are the two soft-fail paths: deferred when condition (iv) cannot yet be established (e.g. an absolute discrete-divergence relation whose reference field already carries non-negligible divergence), and downgraded to OOD-stress when conditions (ii)–(iii) hold only approximately.

3.4. MR card and executable asset format

A retained MR is represented as an MR card recording: identifier, source category, physical or software basis, source-case schema, follow-up transformation, preconditions, boundary-condition compatibility, output mapping, metric, tolerance and provenance, expected verdict classes, exclusion rules, artifact schema, and fault interpretation. Initial card skeletons for the cylinder-flow study are listed with their verdicts in Tables 4 and 5.

Table 1: Domain-validity rubric for candidate MRs.

Criterion	Question asked before retention
Physical basis	What equation, boundary condition, nondimensional law, representation property, numerical assumption, or implementation contract justifies the relation?
Transformation preconditions	What exactly changes from source to follow-up, and what must be preserved?
Boundary-condition compatibility	Does the transformation preserve the intended physical meaning of the domain and boundaries?
Output mapping	Can the expected output relation be measured from available SUT outputs?
Metric and tolerance	What metric is used, and where does the tolerance come from?
Failure diagnosability	What can a violation plausibly mean, and what can it not mean?

Executable assets are generated from MR cards; each includes transformation code, runner configuration, metric computation, verdict logic, and artifact recording. Figure 2 shows the resulting data flow.

3.5. Worked examples: from candidate relation to executable MR card

These are design-time screening decisions, not empirical findings, and do not report whether any SUT passes or fails. Table 2 summarizes three such decisions, stressing different parts of the method: representation semantics, physical symmetry under boundary conditions, and a constraint measurement whose validity depends on a discrete operator and tolerance provenance.

MR-1: node permutation equivariance.. A permutation π is applied to node order and corresponding edge indices without changing coordinates, fields, or boundary labels. Equivariance requires that the output for the permuted case equals $\pi(Y)$ up to numerical tolerance; the output mapping applies π^{-1} before comparison. This is a representation-level contract: tolerance is derived from deterministic repeat runs or machine precision; a failure points to graph adapter, batching, or message-passing implementation issues, not a Navier–Stokes violation.

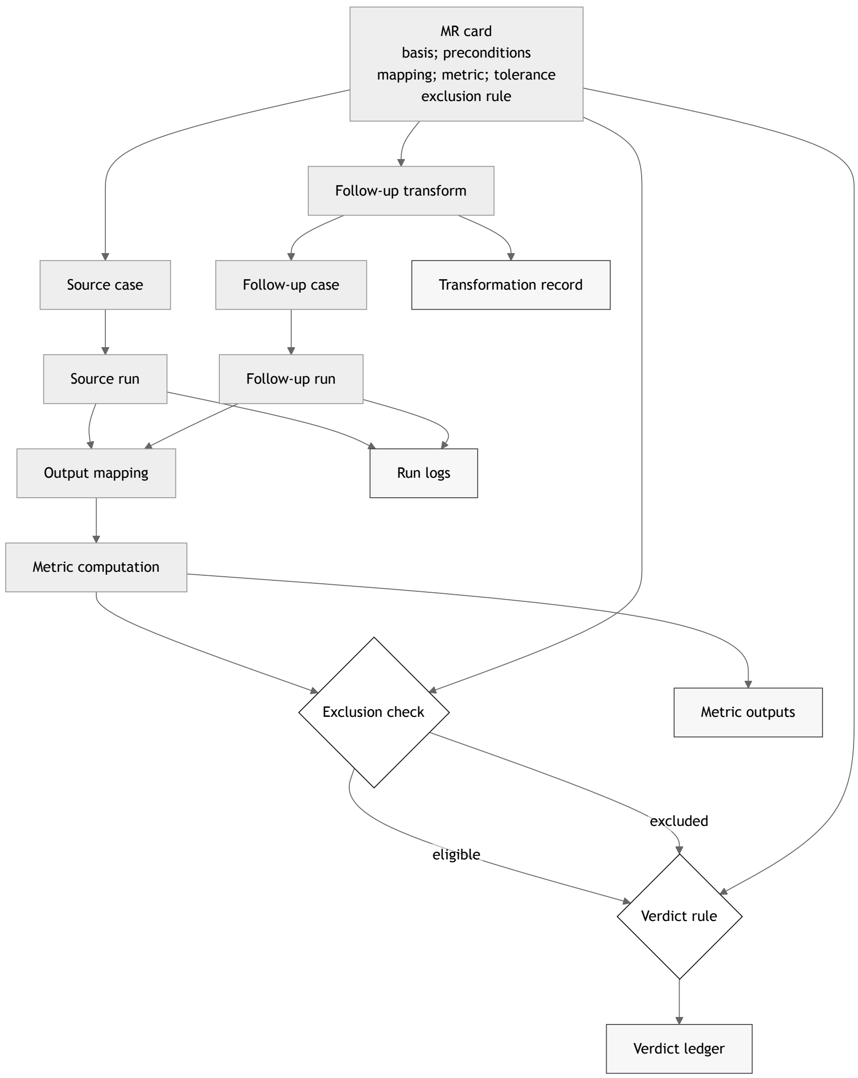

Figure 2: Executable MR asset and verdict data flow. The MR card supplies the basis, preconditions, mapping, metric, tolerance, and exclusion rule; the source and follow-up runs feed the output mapping and metric; the exclusion check and verdict rule emit a relation-level verdict to the ledger together with metric outputs, run logs, and the transformation record as artifacts.

Table 2: Worked screening decisions for three candidate MRs.

Candidate	**Main validity issue**	**Screening decision**	**Reason**
Node permutation equivariance	Graph relabeling must preserve the same physical mesh and fields	Retain as executable representation MR	The transformation changes representation order but not the physical case, provided the adapter and inverse output mapping are deterministic.
Mirror-y equivariance	Geometry, boundary labels, and vector components must transform consistently	Retain only under boundary-compatible conditions	The relation is physically meaningful for symmetric cylinder-flow cases, but invalid if the transformation changes inlet, outlet, wall, or obstacle semantics.
Discrete divergence boundedness	The metric depends on the discrete divergence operator, mesh weights, boundary treatment, and tolerance	Defer or retain as qualified continuity-constraint MR	The physical basis is strong for incompressible flow, but the executable verdict is meaningful only when the numerical operator and threshold are justified.

MR-2: mirror-y equivariance.. Coordinates (x, y) map to $(x, -y)$; the transverse velocity component changes sign; boundary labels transform with the mesh. Retained only when cylinder, inlet, outlet, and wall labels remain semantically compatible after reflection; rejected if the transformation swaps non-equivalent boundaries or leaves vector components unmapped. A violation may indicate a SUT symmetry inconsistency or that the MR was applied outside its domain—the relation-level verdict must preserve that distinction.

MR-3: discrete divergence boundedness.. Predicted velocity fields should remain within a bounded discrete divergence tolerance for eligible interior cells. Retained only after the discrete operator, mesh weights, boundary treatment, and tolerance provenance are specified; the exclusion rule skips cases where boundary treatment dominates the metric or where the flow is outside the incompressibility assumption. A failure is a continuity-constraint violation only after operator and tolerance evidence rule out a numerical artifact.

These examples show how the rubric changes the status of candidate relations: node permutation becomes a retained representation MR, mirror-y a conditional physical-symmetry MR, and divergence a qualified continuity-constraint MR whose verdict depends on numerical instrumentation.

3.6. Relation-level verdicts

An MR execution can produce: **pass** (relation holds within tolerance); **fail** (violated within valid domain); **skip** (precondition missing); **out-of-relation-domain** (transformation produced a case outside the validity domain); **numerical-tolerance issue** (dominated by threshold/resolution uncertainty); or **inconclusive** (insufficient artifact).

These verdicts form a two-dimensional space. One axis is relation-violation magnitude (V/floor ratio); the other is domain-violation magnitude (how far the transformed case lies outside the validity domain). Low domain-violation with high relation-violation is the only region that may be read as SUT inconsistency; high domain-violation is out-of-relation-domain; a violation within the error floor is a numerical-tolerance issue. This decomposition keeps a model-level violation from being confused with a relation applied outside its domain.

In the present study the relation-violation axis is quantitative (mirror-y reports V/floor) but the domain-violation axis is only partially operationalized: the mirror-y case is placed near the validity boundary *qualitatively* because the reflection is non-bijective and the worst node mismatch is about one mesh edge length. A calibrated continuous domain-violation score is left to future work. Each pilot's vertical position is therefore a structural placement, not a calibrated coordinate: the reading is used as an interpretive structure, not as a calibrated boundary measurement. Figure 3 plots the four cylinder-flow pilots in this verdict space; the synthetic-symmetric-mesh probe sits in the SUT-inconsistency region—low domain-violation, high relation-violation—which is what makes the mirror-y finding a model-level result rather than an out-of-domain artefact.

Retained MRs are organized as a three-level hierarchy inspired by MR classification for scientific computing (Yang et al., 2020) — physical-model, computational-model, and code-model relations — used as a predeclared interpretation protocol mapping each MR class to its candidate fault layer. This is a localization protocol, not a validated localization model; Section 5.3 reports a first bounded test against an injected-fault catalogue.

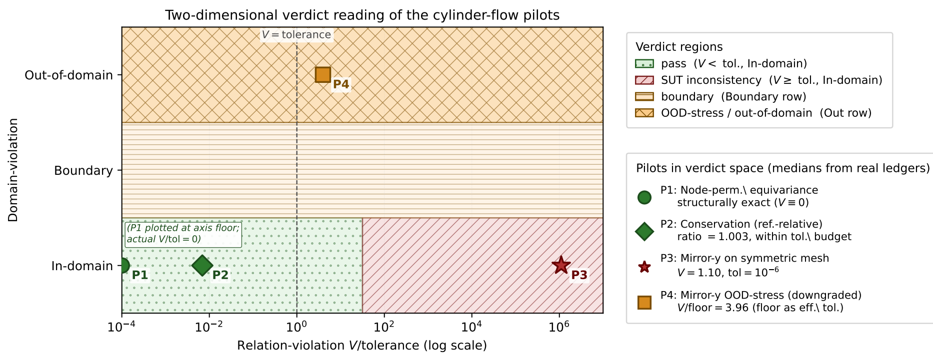

Figure 3: Two-dimensional verdict reading of the cylinder-flow pilots. Horizontal axis: relation-violation V/tolerance on a log scale, where tolerance is the per-MR verdict threshold (machine-precision floor for an exact relation; the reference-relative regression threshold for conservation; the mapping-error floor when an approximate relation has been downgraded to an OOD-stress probe, as for P4). Vertical axis: domain-violation as a three-level categorical bin (In-domain, Boundary, Out-of-domain), kept qualitative because, as Sec. 3.6 notes, it is so far only partially operationalized. The dashed line at $V/\text{tolerance} = 1$ is the pass/fail border for an admissible relation. Coordinates (medians from the metric ledgers; full numbers in Sec. 5.3 and Table 5): P1 node-permutation equivariance $V = 0$ exactly (structurally exact, plotted at the log-axis floor); P2 conservation (reference-relative) ratio $= 1.005$, so $V/\text{tolerance} \approx 0.01$ — two orders of magnitude above P1 yet still well inside pass; P3 mirror-y on the synthetic symmetric mesh, where the rubric admits the exact relation, $V = 1.10$ against tolerance 10^{-6}, $V/\text{tolerance} \approx 1.1 \times 10^{6}$ — the SUT-inconsistency region, which is what removes the out-of-relation-domain objection to the mirror-y finding; P4 mirror-y OOD-stress (downgraded), $V/\text{floor} = 3.96$ with the mapping-error floor as effective tolerance, sitting in the Out-of-domain row. Absolute conservation is not plotted: its admissibility predicate (Sec. 3.3) fails because the absolute tolerance does not dominate the P1 operator floor calibrated in Sec. 5.5, so the rubric never admits it into the verdict space.

4. Empirical Design

4.1. Subject systems

The current evidence uses one trained MeshGraphNets-family implementation and checkpoint. The broader protocol names three intended implementation families (an echowve MeshGraphNets PyTorch/PyG implementation; NVIDIA PhysicsNeMo `vortex_shedding_mgn`; DeepMind TF1 MeshGraphNets), but they remain blocked until matched artifacts exist. Because these SUTs share a task and architecture family, even a completed cross-SUT package would be framed as a same-family stress test, not as evidence for all neural fluid surrogates.

4.2. Planned MR classes

The initial MR set covers representation, geometric/symmetry, continuity/constraint, nondimensional similarity, numerical stability, and temporal/implementation MRs. Time reversal is excluded for viscous cylinder flow; divergence is treated as a continuity constraint, not a Noether-derived conservation law.

4.3. Baselines and comparators

The study uses four comparator families: (1) **Expert MR design**—the primary comparator, making expert domain reasoning explicit and executable; (2) **Generic MR-generation scope contrast**—showing where domain-validity information is needed for SciML; (3) **LLM-assisted candidate generation**—an LLM generates candidates from SUT descriptions; it does not judge MR validity; and (4) **Rollout-accuracy diagnostic**—standard rollout error, reported to assess whether relation-level evidence answers a different validation question. Residual-based or UQ metrics will be reported as diagnostic comparators where feasible, but not treated as MRs unless paired with a source/follow-up transformation and explicit verdict rule.

4.4. RQ-to-evidence map

Table 3 maps each research question to the planned evidence. This table is a pre-results design statement rather than a report of findings.

Table 3: Research questions, efficacy parameters, baselines, and artifacts.

RQ	Primary parameters	Comparators	Artifacts
RQ1	candidate retention rate; rejection reasons; inter-rater agreement	expert MR design; LLM candidates	rubric sheets; candidate ledger; adjudication log
RQ2	executable MR rate; transformation success; asset completeness	expert MR cards; generic scope contrast	MR cards; transformation code; runner configs
RQ3	verdict distribution; out-of-domain rate; interpretation yield	residual/UQ diagnostics where available	verdict ledger; metric outputs; exclusion logs
RQ4	violation rate; violation magnitude; complementarity with rollout error; fault-detection rate if seeded faults exist	expert MR design as primary comparator; LLM/generic candidates as secondary contrasts; rollout accuracy as diagnostic evidence	run logs; plots; confidence intervals; fault ledger

4.5. Efficacy parameters and statistical plan

Primary efficacy parameters are candidate retention rate, executable MR rate, MR violation rate, violation magnitude, fault-detection rate, boundary characterization, and interpretation yield. Secondary parameters include MR construction cost, inter-rater agreement, flakiness rate, localization agreement with seeded-fault layers, and complementarity with rollout accuracy.

The current paper reports only one bounded within-SUT frame-level OOD-stress rate for mirror-y. Broader violation-rate estimates and paired comparisons remain blocked until matched artifacts exist. For the K=6 roster we report paired bootstrap and Wilson 95% intervals, and, following the Empirical Standards, a rank-based effect size (Cliff's δ) with an exact Wilcoxon signed-rank test (Section 5.4), foregrounding effect sizes over significance at this small n. When the blocked comparators are produced, binary verdict comparisons should use McNemar or Fisher-style tests and multiple-comparison correction must be reported.

4.6. Ethics, integrity, and reproducibility

The study does not involve human subjects or personal data. All SUT versions, datasets, checkpoints, scripts, and run logs are recorded when licensing permits; failed, skipped, and inconclusive cases remain in the experiment ledger. LLM use is restricted to candidate generation and evidence organization—not validity judgment. No candidate MR is described as valid without a passing rubric decision and MR card; no OOD violation is described as a program fault without supporting verdict evidence. Third-party artifacts are used according to their licenses; any non-redistributable artifact is described with access instructions.

5. Results

Full cross-SUT, comparative, and fault-detection results remain blocked. This section reports only strictly scoped, within-SUT pilot evidence whose artifacts are committed and validated.

5.1. Claim-to-evidence map

The `PC#` identifiers are paper-level claims, kept distinct from the authoritative runtime claim ledger (`claim-ledger.yml`, claims `C1`–`C9`) so the two cannot be confused. The runtime claim ledger is the single source of truth for runtime-evidence claims; the mapping is PC1→C4, PC2→C1/C4, PC3→C3, PC4→C2, PC5→C7, PC6→C6, PC7→ no runtime-evidence claim (a method/ethics boundary), PC8→C8, PC9→C9, PC10→C10, PC11→C11, PC12→C12, and PC13→C13.

5.2. MR-card-to-verdict map

5.3. Within-SUT pilot evidence

All pilots run on one real trained MeshGraphNets cylinder-flow surrogate, using the read-only Minimum-MR-SubSet checkpoint recorded in the experiment ledger. They exercise three different rubric outcomes and are scoped accordingly; raw outputs, manifests, and metric ledgers are committed under `research_assets/runs/`.

Representation MR. Node-permutation equivariance holds to machine precision (relative L2 = 0.0 at tolerance 1e-6). This is a structural property of message-passing and is reported only as a pipeline sanity check, not as model capability.

Geometric MR. The rubric classifies exact mirror-y equivariance as out-of-relation-domain for this mesh: the reflection is non-bijective, the worst reflected-node mismatch is about one mesh edge length, and the cylinder is off-centre by 7.2 mm. The relation is therefore downgraded to an approximate nearest-neighbour OOD-stress probe scored by the predeclared MR-card metric against a same-space mapping-error floor. Within the same SUT and checkpoint, the approximate mirror-y OOD-stress probe failed on 10 of 10 recorded eval frames (median relative L2 0.737, median V/floor 3.96, 3.02–5.55x mapping-error floor range); every recorded frame is kept in the denominator and none was skipped or inconclusive. This is a bounded within-SUT frame-level OOD-stress result under an approximate reflection: one SUT, one checkpoint, one MR, one eval trajectory. It is not an exact mirror-symmetry result and not a reliability, accuracy, baseline, multi-SUT, exact-symmetry, or geometry-independent claim. The 10 frames are consecutive states of one trajectory and are not independent samples; an exact binomial 95% interval for 10 successes in 10 trials is wide ([0.69, 1.00]). The V/floor ratio is normalized by a mapping-error floor that is itself a function of the same geometric mismatch that triggered the downgrade, so on this asymmetric mesh V/floor cannot fully separate a model-level violation from an amplified geometric artifact; the exact-symmetry run on a symmetric mesh below resolves that ambiguity.

Continuity MR. A P1 discrete-divergence operator yields a non-negligible divergence even for the ground-truth field on this coarse mesh, so an absolute divergence-free tolerance is not calibratable and the absolute mass-conservation relation stays deferred. As a reference-relative diagnostic, the surrogate's predicted next-state divergence stays within about 0.4–0.8% of the reference on the recorded eval frames; the interior-only ratio confirms this is not only a boundary-imposition artifact. This asserts no absolute conservation. Two caveats bound the diagnostic: the reference divergence is not yet decomposed into discrete-operator error, solver projection artifact, or genuinely non-solenoidal training data, so if the reference field is itself materially non-solenoidal the reference-relative ratio is a non-regression guard rather than a conservation measurement; and the diagnostic uses a 50% regression threshold (ratio > 1.5) on two eval frames only, so "passes" means "does not regress conservation by more than 50% on those frames," not "conserves mass."

Rollout-accuracy diagnostic. On the same eval trajectory, the surrogate's one-step next-state prediction error ($v_{\text{pred}} = v_t +$ denormalized predicted delta, the trainer's own convention) has median relative L2 0.0216 (mean 0.044; min 0.0116, max 0.0788) over the nine recorded transitions. The per-transition error is bimodal (alternating $\sim$0.012 and $\sim$0.075), consistent with the vortex-shedding cycle, so we quote both statistics. The surrogate is accurate in-distribution to a few percent per step, yet the mirror-y OOD-stress violation (median 0.737) is larger by roughly an order of magnitude — about 34x the median accuracy and 17x the mean. Both quantities are dimensionless relative L2 but of different objects (equivariance of the model output versus next-state velocity error against the ground truth), so the ratio is an order-of-magnitude gap rather than a precise factor, and we do not compare the symmetric-mesh 1.10 to this number directly because their reference distributions differ. This is the first within-SUT evidence that the relation-level diagnostic and ordinary rollout accuracy answer different validation questions on this surrogate; it is a same-SUT accuracy diagnostic, not a baseline-superiority, multi-trajectory, or cross-SUT claim, and it is one-step, not a free-running rollout-stability result.

Exact mirror-y on a symmetric mesh. To test whether the mirror-y finding survives once the exact relation is admissible, we built a synthetic structured channel mesh that is provably symmetric about the centerline: the reflection is a verified bijection (node-type match 1.0, reflection offset $< 10^{-12}$, edge set invariant), so the admissibility predicate retains the exact relation rather than downgrading it. On one constructed input state the surrogate violated exact mirror-y equivariance with relative L2 1.10 (verdict fail). Because equivariance is oracle-free and structural — a mirror-equivariant model would satisfy it to machine precision regardless of accuracy — the nonzero result is not a geometric artifact, and it is an out-of-sample check that the admissibility predicate did not fit to the original three pilots. We ran one control to separate the learned weights from the input normalizer, which is fit to the asymmetric training data and so is not exactly equivariant in the transverse velocity: zeroing the normalizer's transverse-velocity mean (making it exactly equivariant in that component) changes the violation only from 1.1032 to 1.1014, so the normalizer accounts for about 0.2% of it and the violation is dominated by the learned message-passing weights, which carry no equivariance constraint. Two boundaries remain. The mesh is synthetic, has no obstacle, and is out-of-distribution for the cylinder-trained surrogate, so the *magnitude* 1.10 may be amplified by normalization mismatch relative to in-distribution behaviour; the result should be read as confirming that the surrogate carries no exact mirror-y equivariance constraint, while not bounding the in-distribution equivariance violation, rather than as a calibrated in-distribution magnitude. And this is one input on one mesh; it is not an accuracy, reliability, cross-SUT, or geometry-independent claim. The 1.10 magnitude is also larger than the asymmetric-mesh OOD-stress 0.737, which would be paradoxical if the symmetric mesh were the cleaner setting; the more likely reading is that the synthetic no-obstacle channel is itself more aggressively out-of-distribution for the cylinder-trained surrogate (Poiseuille-like profile rather than vortex shedding, no obstacle wake), so the larger magnitude reflects deeper OOD rather than a cleaner equivariance measurement. The two runs therefore answer different questions — admissibility of the relation, and severity of the violation under OOD — and the symmetric-mesh number should be read as a binary equivariance failure on an admissible relation, not as a directly comparable magnitude.

Seeded-fault detection. We re-implemented, in pure numpy/torch from the read-only Minimum-MR-SubSet witness taxonomy, a catalogue of 10 injected pipeline faults across five fault classes (boundary-condition, mesh-adjacency, normalization-scale, temporal-rollout, physical-channel), and used the paper's own MRs as detectors on the model's predicted update. The conservation (continuity) MR detected the two boundary-condition faults and the gross normalization fault (divergence ratio 3.8–10.6 vs the 1.5 threshold); the mirror-y (symmetry) MR detected a physical-channel and a mesh-adjacency fault (violation rising 69–142% above its 0.735 baseline); node-permutation equivariance detected none, because these faults preserve node-relabeling invariance and the MR stays exact by design. Five of ten mutants were detected by at least one MR (at least one mutant per detected fault class), and the detections localize by MR class — continuity to boundary/scale faults, symmetry to physical-channel/adjacency faults — which is a first bounded test of the interpretation protocol of Section 3.6, suggestive rather than a validation. Three honesty notes bound this. First, the detected faults are gross corruptions (zeroed inflow, non-zero wall, un-denormalized update, swapped velocity channels, permuted edges) that any divergence- or symmetry-sensitive detector would catch; the catalogue is an independent taxonomy, not adversarial to these MRs but not designed for them. Second, two undetected cases are near or by-construction: the edge-drop fault is a near-miss (32% mirror-y change vs the 50% threshold), and the boundary-condition faults are invisible to mirror-y by construction because boundary imposition happens downstream of the update the symmetry MR scores. Third, the remaining undetected faults (doubling a small update, sign-flipping the step, zeroing the transverse update) shift the absolute output without crossing the scored-quantity thresholds, delimiting where these MRs are structurally insensitive. It is one SUT, one checkpoint, one injected-fault catalogue; not a real-world or general fault-detection rate, a reliability claim, or a baseline-superiority claim.

5.4. Multi-checkpoint replication (K=6 checkpoints, one architecture family / one dataset)

To address a single-checkpoint objection without overclaiming a cross-SUT result, the five within-SUT measurements above are replicated across a roster of K=6 MeshGraphNets cylinder-flow checkpoints trained on the same DeepMind dataset using the same read-only Minimum-MR-SubSet trainer: four base-config seed replicas S0–S3 (hidden=64, num_layers=4, training seeds $\{0, 1, 2, 3\}$, 1500 stage-A optimization steps each; S0 reuses the original node-permutation pilot checkpoint and S1–S3 are freshly trained seed replicas) and two configuration variants S4 (hidden=128, seed 0) and S5 (num_layers=6, seed 0). Each checkpoint has a distinct sha256, recorded in its per-SUT manifest and the E1 aggregate file. The same six MR/diagnostic runners are executed against every checkpoint by the E1 orchestrator; the seed-replica family carries a bootstrap 95% confidence interval (B=2000 over S0–S3) and S4, S5 are reported descriptively as single-instance configuration variants. The honesty boundary is explicit: K=6 is a cross-checkpoint / cross-configuration replication across one architecture family and one dataset, not a cross-SUT (cross-architecture-family) generalization, and the seed-replica family standardises only the training seed and so should not be treated as an independent SUT sample. The replication results:

Node-permutation equivariance holds to machine precision (relative L2 = 0.0) on every one of the six checkpoints, confirming the representation-MR sanity check is a structural property of the message-passing pipeline rather than an artifact of one trained instance. It is therefore a pipeline-implementation sanity check—it catches adapter, batching, or graph-encoding bugs that break the network's structural equivariance, not a model-level physical fault—and is read separately from the two model-level (geometry, continuity) MRs.

Mirror-y OOD-stress (approximate reflection on the asymmetric DeepMind mesh) reports a per-checkpoint median relative L2 with seed-family mean 0.774 and 95% bootstrap CI $[0.743, 0.804]$; S4 = 0.764 and S5 = 0.832 sit within or just outside that band. The violation is therefore reproducible across seed and configuration variation, not specific to one checkpoint.

Exact mirror-y on the synthetic symmetric mesh fails on every checkpoint: seed-family mean 1.00, 95% CI $[0.75, 1.15]$ (the wide interval reflects S2's lower 0.63); S4 = 1.10, S5 = 1.00. The exact-symmetry failure is therefore a property of the architecture family, not of one trained instance.

Reference-relative conservation ratio (max over the recorded eval frames) sits tightly above 1: seed-family mean 1.009, 95% CI $[1.007, 1.011]$; S4 $= 1.008$, S5 $= 1.010$. All six checkpoints stay well below the predeclared 1.5 regression threshold, so the reference-relative non-regression verdict reproduces across the family without any checkpoint passing absolute conservation.

One-step rollout accuracy (median relative L2) is also tight: seed-family mean 0.0221, 95% CI $[0.0217, 0.0224]$; S4 $= 0.0226$, S5 $= 0.0221$. The mirror-y OOD-stress violation is therefore about $35\times$ the in-distribution one-step accuracy on *every* checkpoint of the roster, not just on the original pilot. Across the six checkpoints the mirror-y violation exceeds the one-step rollout error on every checkpoint (Cliff's $\delta = 1.0$, complete dominance; exact Wilcoxon signed-rank $p = 0.03125 = 1/32$, the minimum attainable two-sided value at $n = 6$); the effect size is maximal, and we foreground it over the underpowered significance value.

This replication is a robustness check, not a new measurement: the mirror-y violation magnitude survives seed replication and mild configuration variation, the conservation diagnostic stays in its narrow band, and the rollout accuracy is consistent. It licenses no cross-architecture-family generalization, reliability claim, or real-world fault-detection rate.

5.5. Operator-floor resolution sweep (calibration of numerical decidability)

The admissibility predicate of Section 3.3 requires the verdict tolerance to dominate the operator's intrinsic error floor (numerical decidability). For the P1 constant-per-cell discrete divergence operator that scores the continuity MR, that floor is a property of the operator and the mesh, not of any SUT, and it admits a direct empirical calibration. On the same $y = 0$-symmetric structured triangular channel mesh family used by the exact-mirror-y experiment of Section 5.3, at four resolutions $h_0, h_0/2, h_0/4, h_0/8$, we evaluated the analytically divergence-free reference velocity field $u = (\partial_y\psi, -\partial_x\psi)$ with stream function $\psi(x, y) = \sin(\pi x/L_x)\sin(\pi y/L_y)$ and measured the area-weighted RMS of the per-cell discrete divergence. A log–log linear fit of RMS vs. characteristic edge length gives slope **0.988** (all cells, $R^2 = 1.000$) and slope **0.995** (interior-only, $R^2 = 1.000$), matching the theoretical $O(h)$ truncation rate of P1 piecewise-linear shape functions on smooth fields (Figure 4). This empirically calibrates the operator-floor magnitude the verdict tolerance must dominate, confirming the admissibility-predicate gating condition for the continuity MR has a measurable basis. The scope is one mesh family and one smooth analytic field; the calibration confirms the expected rate on this mesh family but does not generalize across mesh geometries.

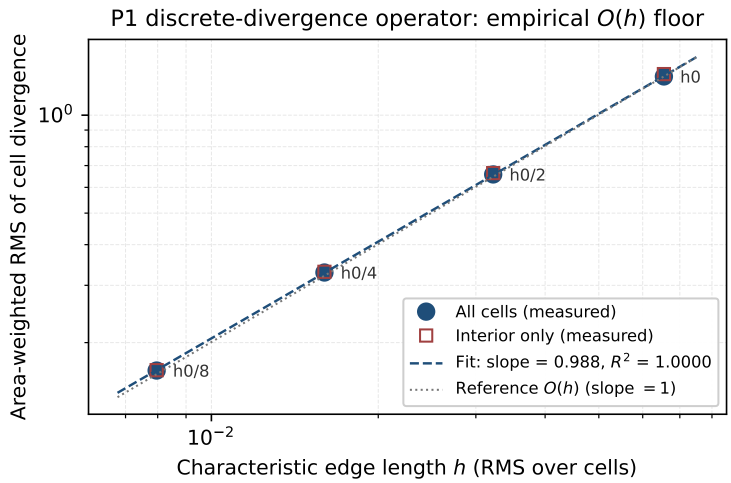

Figure 4: Empirical $O(h)$ convergence of the area-weighted RMS of the P1 discrete-divergence operator on the $y = 0$-symmetric structured triangular channel mesh at four resolutions, evaluated on an analytically divergence-free reference field. All-cell slope = 0.988 ($R^2 = 1.000$); interior-only slope = 0.995. The grey dotted line is the theoretical $O(h)$ reference.

5.6. Fault-detection robustness across the multi-checkpoint roster

To check whether the seeded-fault detection result of Section 5.3 reproduces beyond a single checkpoint and to delimit where its detectors are structurally insensitive, the 10-mutant catalogue is replayed against the K=6 multi-checkpoint roster across five input-permutation seeds (30 trials per mutant per detector), and two predeclared severity dimensions are swept: NS_double_scale $s \in \{1.1, 1.25, 1.5, 2, 4\}$ and PC_zero_vy partial fraction $p \in \{0.25, 0.5, 0.75, 0.85, 0.9, 0.95, 0.99, 1.0\}$ (the canonical mutant is $p = 1.0$). Detection is decided by the same predeclared thresholds as Section 5.3 (node-permutation tolerance 10^{-5}, conservation ratio threshold 1.5, mirror-y relative-change threshold 0.5). Wilson 95% CIs are computed across the (SUT, seed) cells. Raw outputs are committed under `fault-robustness-e3/`.

Per-mutant cross-family detection (R1). Four mutants are detected on every cell (30/30, CI $[0.89, 1.00]$): conservation flags both boundary-condition faults (BC_zero_inflow, BC_nonzero_wall) and the un-denormalized output fault (NS_skip_denorm); mirror-y flags the swapped-channel fault (PC_swap_xy). Five mutants are not detected on any cell (0/30, CI $[0.00, 0.11]$): MA_drop_edges, NS_double_scale, TR_sign_flip, TR_double_step, PC_zero_vy. The by-class localization of the original C10 pilot holds: continuity $\rightarrow$ boundary/scale faults, symmetry $\rightarrow$ physical-channel/adjacency faults, node-permutation $\rightarrow$ none of these mutants (they preserve node-relabeling invariance, and the representation MR stays exact by design). One mutant is configuration-sensitive: mirror-y detects MA_permute_edges on every (S0–S3, seed) cell (20/20 over the seed-replica family) but on neither (S4, seed) nor (S5, seed) cell (0/10 over the wider and deeper configuration variants). The within-family detection rate of that mutant is therefore 20/30 (CI $[0.49, 0.81]$) — a quantitative bound on within-family generalization that is invisible to a single-checkpoint experiment.

NS_double_scale severity sweep (R2). At every multiplicative scale $s \in \{1.1, 1.25, 1.5, 2.0, 4.0\}$ the detection rate stays at 0/6 (Wilson CI $[0.00, 0.39]$). This widens the bounded-insensitivity note from Section 5.3: multiplying the per-step output by a constant factor does not push the reference-relative conservation ratio past 1.5 (the per-step delta is small, so the next-state field stays close enough to the reference), does not move mirror-y by 50% (both the source and the mirrored output scale together, so the relative L2 between them is invariant under output scaling), and does not break node-permutation equivariance (a linear operation is permutation-equivariant). The MR catalogue at the current thresholds is therefore structurally insensitive to multiplicative output scaling on this surrogate across the swept grid, not only at the canonical 2× fault.

PC_zero_vy partial-fraction sweep (R3) — a knife-edge symmetry blind spot. The refined grid resolves *where* detection collapses. At every partial fraction up to $p = 0.99$ the detection rate is 6/6 across the roster (Wilson CI $[0.61, 1.00]$); it falls to 0/6 (CI $[0.00, 0.39]$) only at the exactly-uniform $p = 1.0$. The transition is a step, not a gradual decline — even 1%-asymmetric zeroing is caught on every checkpoint. The responsible detector is *node-permutation*: partial zeroing masks v_y on a node-index-selected subset, which is not relabeling-invariant; the symmetry MR never fires in this sweep (0/6 at all fractions). At $p = 1.0$ the fault becomes simultaneously permutation-invariant and mirror-y-symmetric (uniform $v_y = 0 = -0$ under reflection) and escapes every geometric detector at once (Figure 5). The canonical fault therefore sits in a measure-zero blind region that any partial severity exposes — a within-family finding showing a fault can be invisible to a geometric MR suite precisely when it shares the suite's invariances.

Aggregate reading. The 5-of-10 union detection rate from C10 reproduces across S0–S3 and drops to 4/10 on S4/S5 because MA_permute_edges crosses below the mirror-y threshold there. By-class localization (continuity → boundary/scale; symmetry → physical-channel/adjacency) replicates across the roster. The insensitivity regions — multiplicative output scaling and the PC_zero_vy / adversarial-mutant blind subspace — are bounded with Wilson 95% CIs. This is a within-family robustness result, not a real-world fault-detection rate, cross-architecture generalization, or baseline-superiority claim.

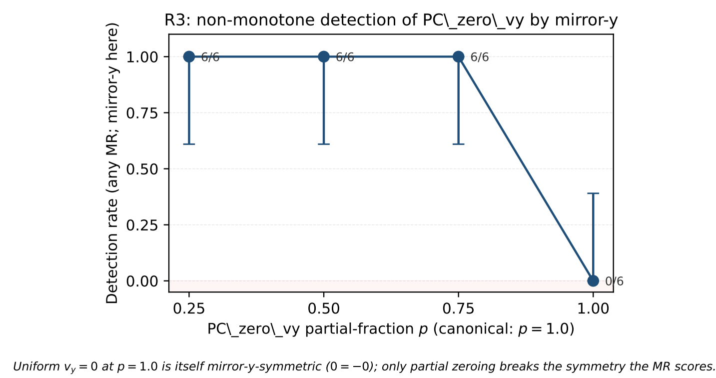

Figure 5: R3 knife-edge detection of PC_zero_vy by the union of node-permutation, mirror-y, and conservation detectors. The any-detector rate is 6/6 across the K=6 roster at every partial fraction up to $p = 0.99$ and drops to 0/6 only at the exactly-uniform $p = 1.0$. The responsible detector is node-permutation (index-selected partial zeroing is not relabeling-invariant), not the symmetry MR (which never fires in this sweep). At $p = 1.0$ the fault is simultaneously permutation-invariant and mirror-symmetric and escapes every geometric detector. Error bars are Wilson 95% CIs across the 6-checkpoint roster.

Adversarial mutants (R4) — the blind region is a subspace, not a point. Two mutants designed to be invariant under all three detectors were replayed on the K=6 roster (`adversarial-mutants-e3-extra/`). A1 adds $c = 0.05$ to δv_x; A2 adds $c \cdot v_y^2$. Both are uniform per-node (permutation-invariant), even under $v_y \mapsto -v_y$ (mirror-y-invariant), and have $\partial_x = 0$ (conservation-invariant). A2 escapes every detector on every checkpoint (any-detector 0/6, Wilson CI $[0.00, 0.39]$; mirror-y relative change $\sim 10^{-4}$, three orders of magnitude below the 0.5 threshold) – the R3 blind point is not isolated. A1 is flagged 6/6 by mirror-y but node-permutation 0/6 and conservation 0/6; A1 trips the mirror-y *magnitude*-shift threshold, not a symmetry break. The blind region under the predeclared thresholds is therefore a subspace whose extent is set by the relative-change tolerances; a calibrated adversary stays inside it. This motivates an explicit symmetry-breaking score in future work; not a real-world fault-detection rate or baseline-superiority claim.

5.7. LLM-generated MR baseline (one-shot, vendor-disjoint panel)

A three-stage vendor-disjoint pipeline (under `llm-mr-baseline/`) executes one LLM-only baseline. Generator gpt-5.5 (bltcy gateway; opus-4-8 rate-limited; temperature = 0; prompt sha256 recorded) proposed $K = 8$ candidate MRs; three vendor-disjoint raters not from the generator's vendor (`glm-5.1`, `kimi-k2.6`, `deepseek-v4-flash`) voted on {valid, borderline, invalid}.

All 8 candidates satisfied the four-condition predicate on their self-declared fields; 7/8 reached panel-majority "valid". Six overlap thematically with this paper's three MRs (2 node-permutation, 1 mirror-y, 4 conservation); the two LLM-only proposals are an implementation-contract pair with no physical basis. The most informative single result is `centerline_reflection_equivariance`, which the predicate admitted but the panel split (valid, borderline, invalid) because the dissenting rater objected that the reflection is not bijective on the asymmetric eval mesh — exactly this paper's reason for downgrading mirror-y to an OOD-stress probe. Inter-rater raw PRA 0.79 / item-unanimous 0.75; Fleiss $\kappa = 0.077$ (the small-sample paradox under near-unanimity).

One generator call, eight candidates, three raters: not a generative-model benchmark or an LLM-vs-method claim. The defensible reading is narrow — the LLM panel proposes mostly the same MR families this paper identifies, agrees on what is admissible, and ratifies through independent reasoning the paper's downgrade of mirror-y.

5.8. Cross-family PINN extension (two bounded points)

To check whether the workflow is MeshGraphNets-specific, two 5×50 MLP PINNs were trained and put through the same three MRs. SUT 1 is 2D viscous Burgers ($\nu = 0.05$, vector field, Dirichlet zero BC, IC symmetric under $y \to -y$); SUT 2 is the 2D heat equation ($\alpha = 0.1$, scalar field, Neumann zero-flux BC, strictly mass-conserving). PDE residual L2 is 1.07 and 1.17×10^{-2}. Checkpoints and FD references are committed under `pinn-cross-family*/`. The MR rewrites parallel Section 5.3. **MR-A** (collocation-point permutation) is exact-by-design for a pointwise MLP (violation 0.0 / 9.9×10^{-9}). **MR-B** (mirror-y equivariance with the PDE residual as the operator floor) gives violation/floor ratio 0.74 (Burgers) and 0.46 (heat) — both pass, the learned symmetry being tighter than the PDE solution. **MR-C** (reference-relative conservation ratio) gives median 1.004 and 0.995 (both pass at the 1.5× gate). Rollout median relative L2 is 1.0 and 1.6×10^{-2}. Magnitudes match the MGN roster (MR-A 0.0; MR-C CI $[1.007, 1.011]$; rollout 0.022), so the 4-condition predicate and the three MR rewrites transfer to two architectures and PDEs (Dirichlet near-conservation and Neumann strict conservation) without re-engineering. Two PINNs, one seed each: two bounded points on the cross-architecture-family applicability map, not a PINN-vs-MGN benchmark.

5.9. Still blocked

The following remain blocked and must not be written as results: cross-SUT or geometry-independent pass/fail rates; comparative superiority over any baseline; general or real-world fault-detection rates; localization accuracy as a validated model; runtime or performance claims; and any claim that one SUT is more reliable than another. The three METBENCH-planned SUTs, and the expert-MR and generic-MR-generation baselines, stay blocked pending their artifacts. Three exceptions are now executed on the existing SUT and reported as scoped diagnostics, not as defeated competitors or general rates: the rollout-accuracy comparator (Section 5.3); a bounded seeded-fault detection result over one 10-mutant catalogue (Section 5.3), whose by-class localization is suggestive evidence for, not a validation of, the interpretation protocol; and the one-shot, vendor-disjoint LLM-MR baseline of Section 5.7. The multi-checkpoint replication of Section 5.4 is a cross-checkpoint / cross-configuration result over one architecture family and one dataset; it does not lift the cross-SUT block.

6. Discussion

6.1. Interpretation of the scoped evidence

The value of the current study is not that every MR finds a new fault beyond rollout accuracy. Retained MRs provide relation-level evidence under explicit transformations; when a violation or deferral occurs, the MR card and verdict rule help distinguish model inconsistency, relation-domain boundary, numerical tolerance issue, and inconclusive evidence.

A natural objection is that the admissibility predicate merely re-describes the three outcomes it was introduced with. Three runs answer this. The symmetric-mesh run uses the predicate out-of-sample and finds a genuine equivariance violation (rel L2 1.10) rather than a pre-arranged result. The rollout-accuracy run shows the mirror-y violation is about 34 times the in-distribution one-step error, so the relation evidence is not a restatement of accuracy. The PINN extension (Section 5.8) transfers the predicate to two further architectures and PDEs at comparable magnitudes, so the framing is not MGN-specific. This is still bounded evidence, not a reliability or geometry-independent claim.

The expert-MR and generic-MR baselines remain blocked; the LLM-baseline of Section 5.7 is a first bounded comparator. The rubric averted two misreadings: without the numerical-decidability gate, absolute discrete-divergence would have been recorded as a conservation *pass* despite the reference field's non-negligible divergence; without the bijectivity check, the asymmetric-mesh mirror-y failure would have been read as a model fault rather than a geometric artifact.

6.2. Boundary of claims

The paper avoids four overclaims: that MRs always beat rollout accuracy; that NOETHER proves the physical validity of cylinder-flow MRs; that LLMs can automatically identify valid MRs; and that one MeshGraphNets family generalizes to all SciML surrogates. The safer claim is that a domain-validity-aware workflow makes MR identification and execution more auditable for a concrete class of SciML SUTs.

6.3. Implications for SciML testing

The scoped evidence shows how SciML testing can move from implicit expert checks to explicit MR assets that complement residuals, UQ, and accuracy by making transformation assumptions and verdict rules inspectable — especially for OOD validation, where the relation's boundary is often as important as the violation. The workflow targets a relation-indexed applicability map in relation space rather than residual space. We do not claim a completed applicability map in this paper; the present evidence is one bounded within-SUT point on such a map, and assembling a calibrated map across SUTs, trajectories, and domain-violation scores remains future work.

7. Threats to Validity

We classify threats following Verdecchia et al. (Verdecchia et al., 2023) and report against the Empirical Standards for software engineering research (Ralph et al., 2021).

Construct validity. MR validity depends on the rubric, physical assumptions, BC compatibility, and tolerances; incorrect discrete operators or thresholds may create false violations/passes. The seeded-fault catalogue is author-implemented from the read-only witness taxonomy, so experimenter bias applies: detection rates do not bound real-world fault-detection effectiveness.

Internal validity. SUT setup, checkpoint differences, random seeds, mesh preprocessing, and runtime nondeterminism may affect verdicts. The experiment ledger records these details.

External validity. The evidence covers one MeshGraphNets family (K=6 roster; Section 5.4) plus two PINNs on 2D Burgers and 2D heat (Section 5.8). The workflow transfers to both PINNs at comparable magnitudes, but three SUTs are not a cross-family generalization: neural operators, other mesh-based simulators, and further PINN architectures remain blocked.

Baseline fairness. Generic MR-generation and LLM baselines may not be designed for SciML. They should be interpreted as scope contrasts and candidate-generation comparators, not as defeated competitors.

Conclusion validity. Small sample sizes, multiple MRs, and many transformation bins can produce unstable conclusions. The mirror-y evidence has a bounded within-SUT frame-level OOD-stress rate, while conservation and node permutation remain pilots. Broader rates, external validity, seeded-fault effectiveness, reliability, accuracy, and baseline claims remain blocked.

Reproducibility. Some SUTs may depend on old runtimes or non-redistributable checkpoints. The paper should disclose such barriers and provide the most complete runnable package possible.

8. Conclusion

This paper presents domain-validity-gated MR identification as an auditable oracle-free testing workflow for SciML surrogates. The scoped pilots

(Section 5.3) replicate across a K=6 MGN roster (Section 5.4) with tight Wilson and bootstrap 95% CIs and transfer to two PINNs (2D Burgers and 2D heat; Section 5.8) at comparable magnitudes; the operator floor is empirically $O(h)$ (Section 5.5); and severity / adversarial sweeps bound where the MR set is structurally insensitive (Section 5.6). It does not support broad cross-architecture generalization, real-world fault-detection rates, baseline superiority, reliability, accuracy, or absolute conservation claims. The central claim is methodological: physically meaningful SciML MRs require explicit validity conditions, executable assets, raw evidence records, and relation-level verdicts.

CRediT authorship contribution statement

Meng Li: Conceptualization, Methodology, Software, Writing – original draft. **Xiaohua Yang:** Supervision, Formal analysis, Writing – review & editing. **Jie Liu:** Investigation, Validation. **Shiyu Yan:** Data curation, Visualization.

Declaration of competing interest

The authors declare no conflict of interest.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used generative AI assistants for code scaffolding, prose copy-editing, and reference cross-checking. After using these tools the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. No generative AI was used to fabricate, alter, or invent experimental data, results, or citations; every numerical claim is backed by a tracked artifact under `research_assets/runs/` and a claim-ledger entry.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (NSFC) General Program (grant no. 12575176), the Hunan Provincial Education Department Project, China (grant no. 202502000728), the Natural Science Foundation of Hunan Province, China (grant no. 2025JJ70193), and an industry-funded research project (grant no. 230KHX060001).

Data availability

The manuscript source, executable test assets (MR cards, runners, rubrics), claim ledger, and raw experiment outputs (checkpoints, per-run `.npy` arrays, metric ledgers) are available in the accompanying replication package. The trained MeshGraphNets cylinder-flow checkpoints derive from the public DeepMind cylinder-flow benchmark; no proprietary data was used.

References

Baral, S., Lee, Y., Khanal, S., Jeon, J., 2026. Xrepit: A deep learning-computational fluid dynamics hybrid framework implemented in openfoam for fast, robust, and scalable unsteady simulations. Computers & Fluids 314, 107075. doi:10.1016/j.compfluid.2026.107075.

Barr, E.T., Harman, M., McMinn, P., Shahbaz, M., Yoo, S., 2015. The oracle problem in software testing: A survey. IEEE Transactions on Software Engineering 41, 507–525. doi:10.1109/TSE.2014.2372785.

Chen, T.Y., Cheung, S.C., Yiu, S.M., 1998. Metamorphic Testing: A New Approach for Generating Next Test Cases. Technical Report HKUST-CS98-01. The Hong Kong University of Science and Technology.

Chen, T.Y., Kuo, F.C., Liu, H., Poon, P.L., Towey, D., Tse, T.H., Zhou, Z.Q., 2018. Metamorphic testing: A review of challenges and opportunities. ACM Computing Surveys 51, 4:1–4:27. doi:10.1145/3143561.

Duque-Torres, A., Pfahl, D., Klammer, C., Fischer, S., 2023a. Bug or not bug? analysing the reasons behind metamorphic relation violations, in: Proceedings of the 2023 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER). doi:10.1109/SANER56733.2023.00080.

Duque-Torres, A., Pfahl, D., Klammer, C., Fischer, S., 2023b. Exploring a test data-driven method for selecting and constraining metamorphic relations. doi:10.48550/arXiv.2307.15522, arXiv:2307.15522.

Duque-Torres, A., Pfahl, D., Klammer, C., Fischer, S., 2023c. Towards a complete metamorphic testing pipeline. doi:10.48550/arXiv.2310.00338, arXiv:2310.00338.

Eniser, H.F., Gros, T.P., Wüstholz, V., Hoffmann, J., Christakis, M., 2022. Metamorphic relations via relaxations: An approach to obtain oracles for action-policy testing, in: Proceedings of the 31st ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), pp. 52–63. doi:10.1145/3533767.3534392.

Gopakumar, V., Gray, A., Zanisi, L., Nunn, T., Pamela, S., Giles, D., Kusner, M.J., Deisenroth, M.P., 2025. Calibrated physics-informed uncertainty quantification, in: Proceedings of the 42nd International Conference on Machine Learning (ICML). doi:10.48550/arXiv.2502.04406, arXiv:2502.04406.

Hiremath, D.J., Claus, M., Hasselbring, W., Rath, W., 2021. Towards automated metamorphic test identification for ocean system models, in: 2021 IEEE/ACM 6th International Workshop on Metamorphic Testing, pp. 31–35. doi:10.1109/MET52542.2021.00014, arXiv:2103.09782.

Kanewala, U., Bieman, J.M., 2019. Metamorphic testing of scientific software: A machine learning approach. Journal of Software: Evolution and Process 31, e1894. doi:10.1002/smr.1894.

Kanewala, U., Bieman, J.M., Ben-Hur, A., 2016. Predicting metamorphic relations for testing scientific software: A machine learning approach using graph kernels. Software Testing, Verification and Reliability 26, 245–269. doi:10.1002/stvr.1594.

Karniadakis, G.E., Kevrekidis, I.G., Lu, L., Perdikaris, P., Wang, S., Yang, L., 2021. Physics-informed machine learning. Nature Reviews Physics 3, 422–440. doi:10.1038/s42254-021-00314-5.

Krishnapriyan, A.S., Gholami, A., Zhe, S., Kirby, R.M., Mahoney, M.W., 2021. Characterizing possible failure modes in physics-informed neural networks, in: Advances in Neural Information Processing Systems, pp. 26548–26560.

Li, M., Yang, X., Liu, J., Yan, S., 2026. Noether: A constructive framework for metamorphic pattern discovery from operator algebras. doi:10.48550/arXiv.2605.17390, arXiv:2605.17390.

Li, Z., Kovachki, N., Azizzadenesheli, K., Liu, B., Bhattacharya, K., Stuart, A., Anandkumar, A., 2021. Fourier neural operator for parametric partial differential equations, in: International Conference on Learning Representations.

Lin, Q., Kuo, F.C., Liu, H., Poon, P.L., Chen, T.Y., Tse, T.H., 2020. Exploratory metamorphic testing for scientific software. Computing in Science and Engineering 22, 78–89. doi:10.1109/MCSE.2018.2880577.

Mandrioli, C., Shin, S.Y., Bianculli, D., Briand, L., 2025. Testing cps with design assumptions-based metamorphic relations and genetic programming. IEEE Transactions on Software Engineering 51, 1666–1684. doi:10.1109/TSE.2025.3563121.

Olsen, P.C., Raunak, M.S., Rothermel, G., 2019. Increasing validity and reliability in simulation-based testing. IEEE Transactions on Reliability 68, 1322–1337. doi:10.1109/TR.2019.2906504.

Pfaff, T., Fortunato, M., Sanchez-Gonzalez, A., Battaglia, P.W., 2021. Learning mesh-based simulation with graph networks, in: International Conference on Learning Representations. doi:10.48550/arXiv.2010.03409, arXiv:2010.03409.

Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. Journal of Computational Physics 378, 686–707. doi:10.1016/j.jcp.2018.10.045.

Ralph, P., et al., 2021. Empirical standards for software engineering research. ACM SIGSOFT. https://github.com/acmsigsoft/EmpiricalStandards.

Raunak, M.S., Olsen, P.C., Simko, G., Kuhn, D.R., 2021. A continuum of oracles for testing scientific software, in: 2021 IEEE/ACM 6th International Workshop on Metamorphic Testing, pp. 18–25. doi:10.1109/MET52542.2021.00015.

Reichert, M., Bouaziz, L., Verbeke, B., Eberle, C., Kratzert, F., Klotz, D., Gauch, M., Schulz, K., Hofmann, T., Holzleitner, M., Klambauer, G., Hochreiter, S., Nearing, G., Gnann, S., 2024. Metamorphic testing of machine learning and conceptual hydrologic models. Hydrology and Earth System Sciences 28, 2505–2529. doi:10.5194/hess-28-2505-2024.

Segura, S., Fraser, G., Sanchez, A.B., Ruiz-Cortes, A., 2016. A survey on metamorphic testing. IEEE Transactions on Software Engineering 42, 805–824. doi:10.1109/TSE.2016.2532875.

Verdecchia, R., Engström, E., Lago, P., Runeson, P., Song, Q., 2023. Threats to validity in software engineering research: A critical reflection. Information and Software Technology 164, 107329. doi:10.1016/j.infsof.2023.107329.

Wang, S., Teng, Y., Perdikaris, P., 2021. Understanding and mitigating gradient flow pathologies in physics-informed neural networks. SIAM Journal on Scientific Computing 43, A3055–A3081. doi:10.1137/20M1318043.

Wang, W., Hakimzadeh, M., Ruan, H., Goswami, S., 2025. Time-marching neural operator-finite element coupling: Ai-accelerated physics modeling. Computer Methods in Applied Mechanics and Engineering 446, 118319. doi:10.1016/j.cma.2025.118319.

Xie, X., Ho, J.W.K., Murphy, C., Kaiser, G., Xu, B., Chen, T.Y., 2011. Testing and validating machine learning classifiers by metamorphic testing. Journal of Systems and Software 84, 544–558. doi:10.1016/j.jss.2010.11.920.

Yang, F., Zhang, J., Chen, T.Y., Liu, H., 2020. Hierarchical classification model for metamorphic relations of scientific computing programs. Computer Science doi:10.11896/jsjkx.200200015.

Table 4: Compact claim-to-evidence map for the current manuscript.

Claim	**Status**	**Evidence**	**Boundary**
PC1-domain-validity-rubric	Method claim	Rubric asset and manuscript method section	Does not prove physical validity by itself.
PC2-mr-card-executable-assets	Workflow claim	MR cards and validators	Not every card has cross-SUT evidence.
PC3-baseline-comparison-blocked	Blocked	Claim and experiment ledgers	No baseline superiority, seeded-fault, localization, or runtime claim.
PC4-node-permutation-sanity	Observed pilot	Node-permutation metric ledger	One SUT and one pilot case only.
PC5-conservation-diagnostic-deferred	Diagnostic; absolute claim deferred	Conservation metric ledger and report	absolute conservation remains deferred.
PC6-mirror-y-ood-stress	Observed bounded pilot	Mirror-y rate metric ledger and claim ledger	failed on 10 of 10 recorded eval frames; not a reliability, accuracy, baseline, multi-SUT, exact-symmetry, or geometry-independent claim.
PC7-llm-candidate-support-only	Process boundary	Method and ethics sections	LLMs organize candidates; they do not judge MR validity.
PC8-rollout-accuracy-diagnostic	Observed	Rollout-accuracy metric ledger	Same-SUT one-step accuracy (median 0.0216); mirror-y is ∼34x larger; not baseline superiority.
PC9-exact-mirror-y-symmetric-mesh	Observed	Symmetric-mesh metric ledger	Exact relation admissible and fails (rel L2 1.10) on a synthetic OOD mesh; not accuracy/cross-SUT.
PC10-seeded-fault-detection	Observed	Seeded-fault metric ledger	MRs as detectors catch 5/10 injected mutants, localizing by MR class; not a general/real-world fault-detection rate.
PC11-multicheckpoint-replication	Observed	E1 aggregate and per-SUT manifests in `multicheckpoint/`	K=6 checkpoints of one MeshGraphNets family (S0–S3 seed replicas, S4 wider, S5 deeper); cross-checkpoint replication, not cross-architecture-family generalization.
PC12-operator-floor-resolution	Observed	Operator-floor sweep report	Log–log slope 0.988 (all)/0.995 (interior), $R^2 = 1.0$ over four resolutions; calibrates the admissibility predicate, not a mesh-family-independent claim.
PC13-fault-detection-robustness	Observed	E3 robustness report	30-trial Wilson CIs per (mutant, detector) across K=6; NS_double_scale undetected over $\{1.1, 1.25, 1.5, 2, 4\}$; PC_zero_vy knife-edge — 6/6 detection (via node-permutation, not symmetry) at every p up to 0.99 and 0/6 only at the exactly-uniform $p = 1.0$; MA_permute_edges configuration-sensitive. Within-family only.

Table 5: MR-card-to-verdict map for the three executed pilots.

MR card	**Rubric decision**	**Runtime verdict**	**Interpretation boundary**
Node permutation equivariance	Retained as representation MR	pass sanity; relative L2 = 0.0	Supports one executable representation path; does not establish model reliability.
Mirror-y equivariance (asymmetric eval mesh)	Exact relation out-of-relation-domain; downgraded to approximate OOD-stress	fail on 10 of 10 recorded eval frames; median relative L2 0.737; median V/floor 3.96	Bounded within-SUT OOD-stress violation for one trajectory; not by itself exact symmetry, cross-SUT, or geometry-independent evidence.
Mirror-y equivariance (synthetic symmetric mesh)	Exact relation admissible (bijection verified, offset $< 10^{-12}$, type-match 1.0)	fail; relative L2 1.10 on one input state	Exact-symmetry violation where the relation is admissible, removing the out-of-relation-domain objection; synthetic no-obstacle OOD mesh, one input.
Discrete divergence / conservation	Absolute mass-conservation MR deferred; reference-relative diagnostic retained	inconclusive: reference-relative non-regression guard on 2 frames (not scored as a conservation pass)	At 2 frames with a 50% slack threshold and an undecomposed reference divergence, this guards against regression but does not establish conservation.

Table 6: Multi-checkpoint replication aggregates over the K=6 SUT roster. CI is a B=2000 bootstrap over the seed-replica family S0–S3; S4 and S5 are reported as single-instance configuration variants.

MR / diagnostic (per-SUT scalar)	**S0–S3 mean**	**95% CI**	**S4**	**S5**
Node-permutation rel L2	0.0	$[0.0, 0.0]$	0.0	0.0
Mirror-y OOD-stress median rel L2	0.774	$[0.743, 0.804]$	0.764	0.832
Exact mirror-y on sym mesh rel L2	1.00	$[0.75, 1.15]$	1.10	1.00
Conservation max ratio	1.009	$[1.007, 1.011]$	1.008	1.010
One-step rollout median rel L2	0.0221	$[0.0217, 0.0224]$	0.0226	0.0221
Seeded-fault union (k/10)	5/10 (each)	—	4/10	4/10

Table 7: Per-mutant Wilson 95% confidence intervals from the E3 sweep across the K=6 multi-checkpoint roster and 5 input-permutation seeds (30 trials per mutant). "Detecting MR" is the MR whose threshold the mutant trips at its canonical severity.

Mutant	**Fault class**	**Detecting MR**	**k/n**	**Rate**	**95% CI**
BC_zero_inflow	boundary cond.	conservation	30/30	1.00	$[0.89, 1.00]$
BC_nonzero_wall	boundary cond.	conservation	30/30	1.00	$[0.89, 1.00]$
NS_skip_denorm	normalization	conservation	30/30	1.00	$[0.89, 1.00]$
PC_swap_xy	physical channel	mirror-y	30/30	1.00	$[0.89, 1.00]$
MA_permute_edges	mesh adjacency	mirror-y (S0–S3 only)	20/30	0.67	$[0.49, 0.81]$
MA_drop_edges	mesh adjacency	(none)	0/30	0.00	$[0.00, 0.11]$
NS_double_scale	normalization	(none)	0/30	0.00	$[0.00, 0.11]$
TR_sign_flip	temporal rollout	(none)	0/30	0.00	$[0.00, 0.11]$
TR_double_step	temporal rollout	(none)	0/30	0.00	$[0.00, 0.11]$
PC_zero_vy	physical channel	(none at $p = 1.0$)	0/30	0.00	$[0.00, 0.11]$